



Enhanced lithium ion conduction and the size effect on interfacial phase in Li_2ZnI_4 –mesoporous alumina composite electrolyte

Hideki Maekawa ^{a,b,c,*}, Takashi Iwatani ^a, Hangyan Shen ^{a,1},
Tsutomu Yamamura ^a, Junichi Kawamura ^d

^a Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan

^b Center for Interdisciplinary Research, Tohoku University, Sendai 980-8578, Japan

^c CREST, Japan Science and Technology Corporation, Kawaguchi 332-0012, Japan

^d Institute for Multidisciplinary Research, Tohoku University, Sendai, 980-8577, Japan

Received 30 August 2007; received in revised form 17 October 2007; accepted 22 October 2007

Abstract

The enhancement of lithium ion conductivity was observed for the composite type electrolyte based on Li_2ZnI_4 and sol–gel synthesized mesoporous- Al_2O_3 with controlled pore size (2–10 nm). Compositional and pore size dependence of the electrical conductivity was investigated. More than fifty times enhancement of the electrical conductivity was observed for the composite with mesopore size greater than 5.0 nm. ^6Li MAS-NMR measurement was performed for the composite electrolytes based on LiI and Li_2ZnI_4 . Both LiI and Li_2ZnI_4 gave an identical ^6Li NMR peak position at –4.9 ppm. Two additional lithium ion environments, at –0.8 ppm and –2.8 ppm, respectively, were observed for the composite. Comparison with a ^6Li cross polarization MAS-NMR measurement clarified the origin of –0.8 ppm peak as a surface lithium ion adjacent to OH^- groups. The high conductivity of the composite was suggested to originate from new interfacial phase that have ^6Li NMR peak position at –2.8 ppm.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Mesoporous alumina; Lithium zinc iodide; Composite ionic conductor; Enhanced conductivity; Size effect; Size dependence

1. Introduction

Fabrication of ionic conductor composites having mesoscale interface by artificial structure-directing methods is an attractive approach for the development of high performance ionic conductor both in fundamental and application point of view [1,2]. A dispersion of insoluble dielectric oxide particles such as Al_2O_3 in ionic conductors is known to increase total electrical conductivity [3–12]. The enhancement of the conductivity was due to various interfacial phenomena such as a space-charge effect, a lattice deformation and a formation of surface specific phase [3–12]. Previously, our group reported a peculiar en-

hancement of the conductivity for the composite type electrolyte composed of lithium iodide (LiI) and a high surface area and pore size controlled mesoporous- Al_2O_3 [9–12]. Further enhancement of the electrical conductivity ($2.5 \times 10^{-4} \text{ Scm}^{-1}$ at 298 K) compared to the previously known LiI -alumina composites was observed. The enhancement of the conductivity was shown to be dependent on the pore size where the smaller pore size resulted in an increase of the conductivity, except for the smallest pore size (2.8 nm). A formation of the interfacial amorphous phase and its size effect was suggested to be responsible for the enhanced conductivity. However, the structural nature of the interfacial phase and the origin of the pore size dependence were not yet clarified.

In the present investigation, mixed iodide lithium ion conductor, Li_2ZnI_4 , with an olivine type crystal structure [13] was used for the fabrication of the composite with pore size controlled mesoporous- Al_2O_3 . The electrical conductivity and the pore size dependence were measured. The structural investigation

* Corresponding author. Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan. Tel./fax: +81 22 795 3763.

E-mail address: maekawa@material.tohoku.ac.jp (H. Maekawa).

¹ Present address: College of Material Science and Engineering, China Jiliang University, Hangzhou 310018, China.

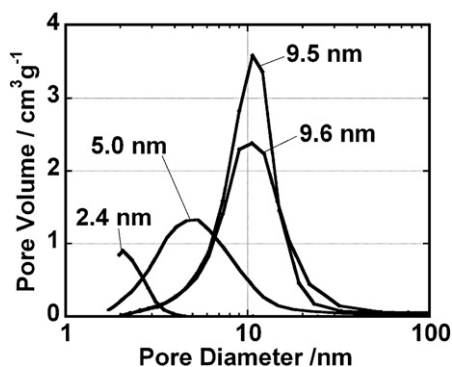


Fig. 1. The pore diameter distribution curve for sol-gel synthesized mesoporous- Al_2O_3 obtained by the BJH method from the adsorption branch of the N_2 sorption isotherm at 77 K.

was performed with a high-resolution ^6Li MAS-NMR spectroscopy. For a comparison purpose, LiI based composites were also investigated. The mesopore size effect on the conductivity and the structure of ionic conductor phase impregnated into mesopore was focused.

2. Experimental

Powdered mesoporous- Al_2O_3 with controlled pore size was prepared by the sol-gel method utilizing neutral surfactant (X114, Aldrich) and cyclohexane as a swelling agent [10,12]. Mesoporous- Al_2O_3 with controlled pore size were heated at 573 K over 24 h under vacuum to remove the adsorbed water, and was kept under a globebox atmosphere (<1 ppm v H_2O). Pore size, pore volume and BET surface area were obtained by N_2 adsorption and desorption isotherms at 77 K by Micrometrics ASAP2010. Crystalline Li_2ZnI_4 was synthesized from LiI (99.999%, Aldrich) and ZnI_2 (99.999%, Aldrich) by a calcination in a sealed Pyrex glass at 473 K, for 30 h. Mesoporous- Al_2O_3 and Li_2ZnI_4 were mixed and pressed into a pellet in a globebox. The pellet was sealed into a Pyrex glass and heated at 753 K for 5 h and calcined at 473 K for 30 h. For a comparison purpose, $\text{LiI} \cdot \text{H}_2\text{O}$ was synthesized from distilled water and anhydrous LiI in a sealed Pyrex glass at 110° C for 3 h.

X-ray diffraction measurement ($\text{Cu K}\alpha$) for the composite sample was performed by polyimide film cell to avoid hydration from the humidity. Temperature dependence of the conductivity was observed from the complex ac impedance spectrum by using Solartron 1260 impedance meter with a home made conductivity cell. ^6Li magic angle spinning (MAS)-NMR spectra were obtained by JEOL ECA-600 spectrometer operated at 88.32 MHz under 14 T field, equipped with single tuned 4 mm ϕ MQ-MAS probe. ^6Li cross polarization (CP) MAS-NMR measurement was operated by Chemagnetics CMX-300 spectrometer at 44.19 MHz under 7 T field. Samples were packed into 4 mm ϕ rotor under a globebox atmosphere. Although ^6Li sensitivity is much lower than commonly used ^7Li , sharp and high-resolution spectrum can be expected due to much smaller nuclear-quadrupole moment for ^6Li . Peak

Table 1

Pore size, BET surface area and pore volume of sol-gel synthesized mesoporous Al_2O_3 used for the fabrication of lithium ion conductor composite

Sample	Pore size/nm	BET surface area/ $\text{m}^2 \text{g}^{-1}$	Pore volume/ $\text{cm}^3 \text{g}^{-1}$
1	2.4	368	0.15
2	5.0	455	0.76
3	9.5	421	1.28
4	9.6	367	1.12

separation and integration of the NMR spectra were performed by the Winfit program [14].

3. Results and discussions

3.1. Fabrication of mesoporous- Al_2O_3

The N_2 adsorption and desorption isotherms of mesoporous- Al_2O_3 were Type IV, with a small hysteresis that indicates a minor deviation from a regular array of cylinders. The pore diameter distribution curve was obtained by the BJH method from the adsorption branch of the N_2 sorption isotherm. As shown in Fig. 1, sharp peak was observed on the BJH pore size distribution, reflecting a well ordered pores. Four kinds of mesoporous- Al_2O_3 with controlled pore size (2.4 nm, 5.0 nm, 9.5 nm and 9.6 nm) were selected for the fabrication of the composite. Samples with similar surface area, ca. 400 m^2/g , were selected for the fabrication of the composite, as listed in Table 1.

3.2. Ionic conductivity

Fig. 2 shows a Cole-Cole plot of 0.46 $\text{LiZn}_{0.5}\text{I}_2$ –0.54 mesoporous- Al_2O_3 (pore size=9.6 nm) obtained by an ac impedance measurement with acetylene black electrodes. A single and slightly stretched semicircle was obtained. Bulk electrical conductivity was obtained from the diameter of the arc. Fig. 3(a) and (b) show a compositional dependence of the ionic conductivity for Li_2ZnI_4 –mesoporous- Al_2O_3 (pore size=9.6 nm) composites. The conductivity increased with an

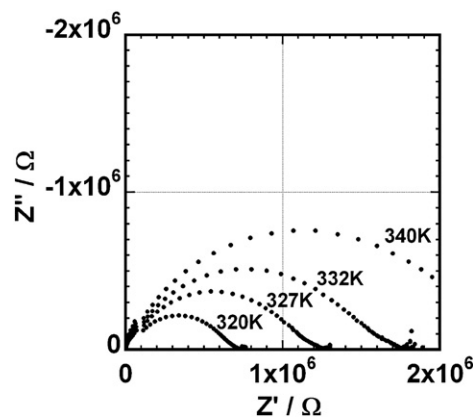


Fig. 2. The temperature variation of Cole-Cole plots for 0.46 $\text{LiZn}_{0.5}\text{I}_2$ –0.54 mesoporous- Al_2O_3 (pore size=9.6 nm) obtained by an ac impedance measurement with acetylene black electrodes. A dc electrical conductivity of the sample was obtained from the diameter of the arc.

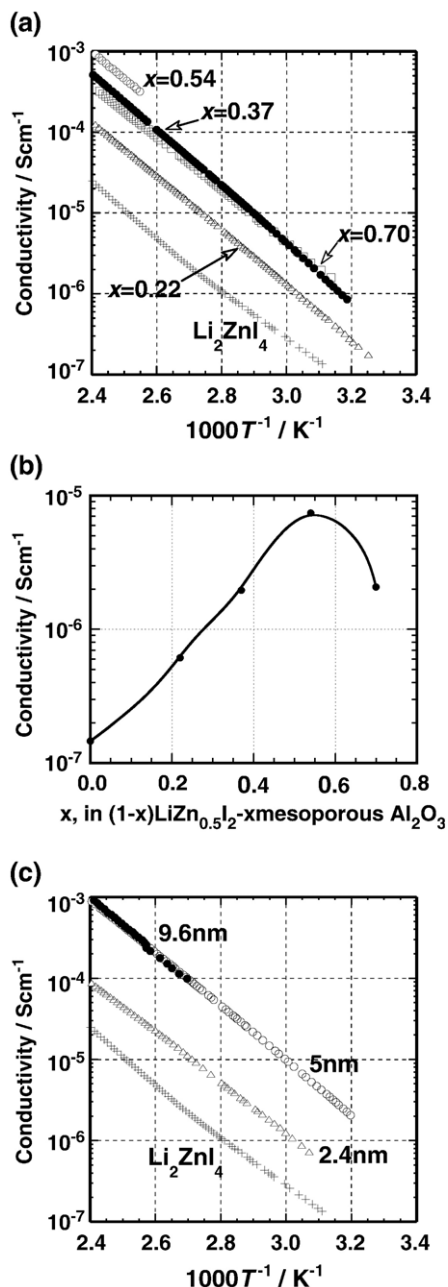


Fig. 3. (a) Arrhenius plots of the dc ionic conductivity for $(1-x)\text{LiZn}_{0.5}\text{I}_2-x$ mesoporous- Al_2O_3 (pore size=9.6 nm) composites. The maximum conductivity, more than fifty times higher than that of pure Li_2ZnI_4 , was obtained for $x=0.54$ composition. (b) Compositional dependence of the dc ionic conductivity for $(1-x)\text{LiZn}_{0.5}\text{I}_2-x$ mesoporous- Al_2O_3 (pore size=9.6 nm) composites at 50°C. (c) Arrhenius plots of the dc ionic conductivity for 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 composite prepared by mesoporous- Al_2O_3 with different pore size.

increase of the mesoporous- Al_2O_3 contents. The maximum conductivity, more than fifty times higher than that of pure Li_2ZnI_4 , was obtained for $x=0.54$ composition.

Fig. 3(c) shows temperature dependence of the electrical conductivity of 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 composite prepared by mesoporous- Al_2O_3 with different pore size. The enhancement was small for 2.4 nm pore, whereas for 5.0 and 9.6 nm pores show similar enhancement effect on the

conductivity. The apparent activation energy for the electrical conductivity was almost constant over composition and pore size variation. This behavior suggests that the interfacial conductivity enhancement is mainly due to the increase of the carrier concentration and the lithium ion mobility was not influenced significantly. The decrease of the contribution of the low conductivity phase to the total electrical conductivity due to an increase of interface specific high ionic conduction phase can also be the reason of the enhancement.

3.3. XRD

XRD patterns for pure Li_2ZnI_4 and 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 composite are shown in Fig. 4(a). The XRD of the composite shows peaks identical to the Li_2ZnI_4 phase with a broadened peak linewidth. Crystallite size of the Li_2ZnI_4 in the composite was estimated to be 20 nm by using Sherrer's equation. Because the estimated crystallite size is much larger than the mesopore size, Li_2ZnI_4 crystals responsible for the XRD pattern should be located outside the mesopore. Thus, the existence of the Li_2ZnI_4 phase impregnated into mesopore could not be observed by XRD. For the characterization of the unknown phase, ^6Li MAS-NMR measurement was conducted.

3.4. ^6Li MAS-NMR

Compositional dependence of the ^6Li MAS-NMR spectra for $(1-x)\text{LiZn}_{0.5}\text{I}_2-x\text{Al}_2\text{O}_3$ composites ($x=0, 0.37, 0.54, 0.70$) are

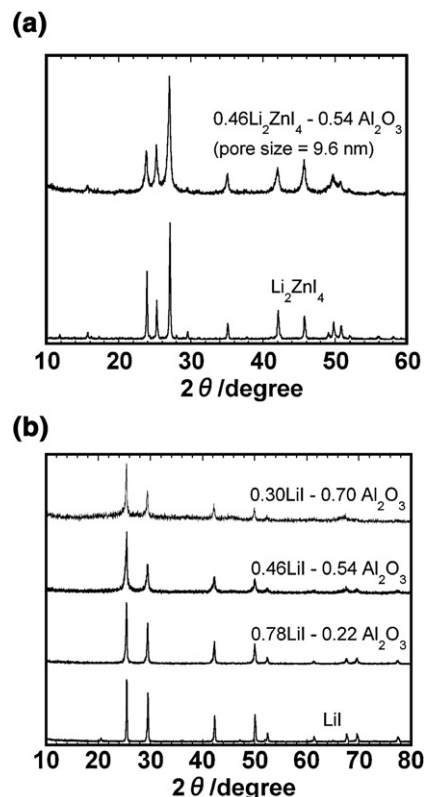


Fig. 4. (a) XRD patterns for pure Li_2ZnI_4 and 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 composite (pore size=9.6 nm). (b) XRD patterns for pure LiI and LiI -mesoporous- Al_2O_3 composite (pore size=9.6 nm) with different composition.

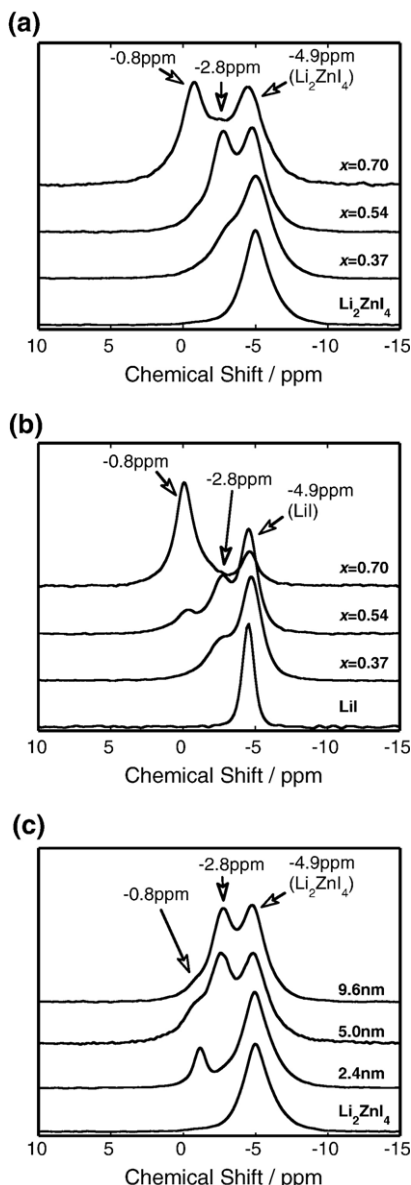


Fig. 5. (a) Compositional dependence of the ^6Li MAS-NMR spectra for $(1-x)\text{LiZn}_{0.5}\text{I}_2-x$ mesoporous- Al_2O_3 (pore size=9.6 nm) composites ($x=0, 0.37, 0.54, 0.70$). A single peak at -4.9 ppm was observed for $\text{LiZn}_{0.5}\text{I}_2$. An additional lithium ion environment at -0.8 ppm and -2.8 ppm, respectively, appeared in the composite. (b) Compositional dependence of the ^6Li MAS-NMR spectra for $(1-x)\text{LiI}-x$ mesoporous- Al_2O_3 (pore size=9.6 nm) composites ($x=0, 0.37, 0.54, 0.70$). (c) ^6Li MAS-NMR spectra for pure $\text{LiZn}_{0.5}\text{I}_2$ and 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 composite with different mesopore sizes. A lack of -2.8 ppm peak was seen for 2.4 nm mesopore composite.

shown in Fig. 5(a). A single peak at -4.9 ppm was observed for Li_2ZnI_4 . An additional lithium ion environment at -0.8 ppm and -2.8 ppm, respectively, appeared in the composite. The -2.8 ppm peak intensity showed a maximum at $x=0.54$, whereas -0.8 ppm peak gradually increased until $x=0.7$. Fig. 5 (b) shows ^6Li MAS-NMR spectra of $(1-x)\text{LiI}-x\text{Al}_2\text{O}_3$ composites ($x=0, 0.37, 0.54, 0.70$). Similar to Li_2ZnI_4 , a single peak at -4.9 ppm was observed for pure LiI . The linewidth of LiI was smaller than Li_2ZnI_4 , and may reflect a higher symmetry of the lithium environment for the LiI rocksalt

structure. An additional lithium ion environment at -0.8 ppm and -2.8 ppm, respectively, was also observed for the LiI composite. It is clear that at least two different lithium environments exist in the amorphous phase in the composite. The multiple lithium environments were also observed on the previous ^7Li MAS-NMR measurement for LiI -nano particle Al_2O_3 composite by Ardel et al. [15]. However, the origin of the multiple peaks has not been clarified.

To assign the origin of two peaks, ^6Li cross polarization (CP) MAS-NMR was conducted. In the CP measurement, ^6Li nucleus adjacent to protons can be observed selectively. A single peak at -0.8 ppm (Fig. 6(b)) was observed for the CP/MAS measurement of the composite. This suggests that the peak at -0.8 ppm originates from the interfacial lithium ion adjacent to the surface OH^- of mesoporous- Al_2O_3 .

The pore size dependence of the ^6Li MAS-NMR spectra for 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 composite is shown in Fig. 5(c). The lineshapes of spectra were similar for 9.6 and 5.0 nm pores, respectively, whereas that of 2.4 nm pore lacks intensity from -2.8 ppm peaks. By a comparison of Figs. 3(c) and 5(a) and (b), qualitative relationships can be seen between the conductivity and the relative intensity of the -2.8 ppm peak. The intensity of -2.8 ppm peak is proportional to the increment of the conductivity. Thus, the origin of -2.8 ppm peak was suggested to be a new phase formed in the mesopore, and plays a major role on the enhanced conductivity. The drastic decrease of the conductivity and the lack of -2.8 ppm peak intensity for

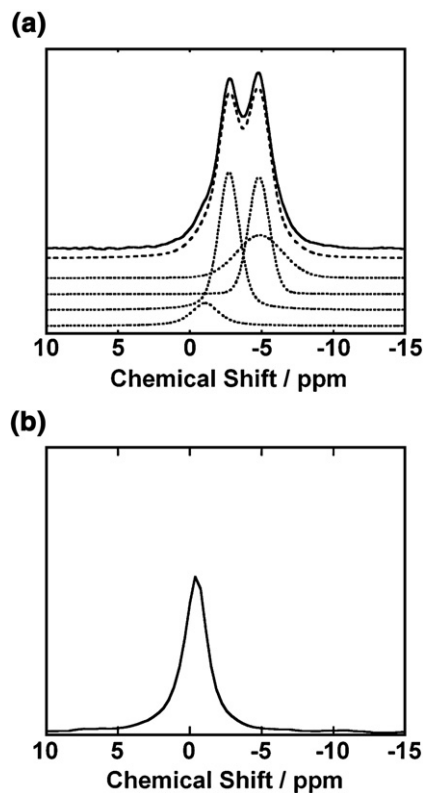


Fig. 6. (a) An example of peak separation shown for the 0.46 $\text{LiZn}_{0.5}\text{I}_2-0.54$ mesoporous- Al_2O_3 (pore size=9.6 nm) composite. (b) ^6Li Cross-polarization (CP) MAS-NMR spectrum for 0.30 $\text{LiZn}_{0.5}\text{I}_2-0.70$ mesoporous- Al_2O_3 composite (pore size=9.6 nm).

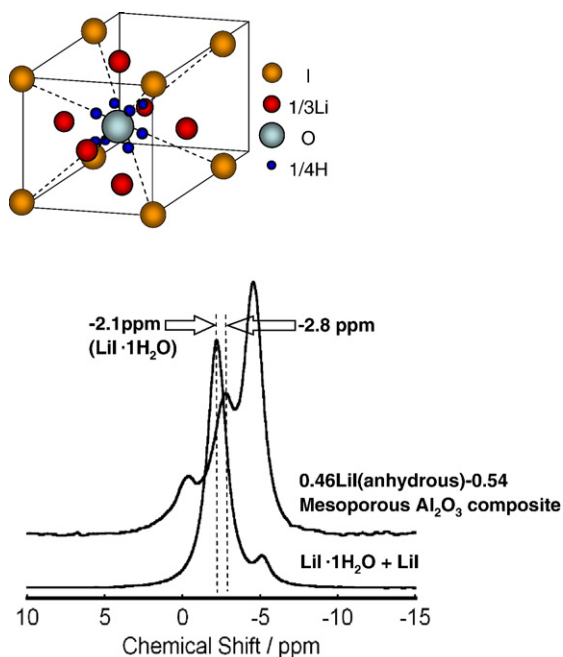


Fig. 7. Comparison of ^6Li NMR spectra of the $\text{LiI}\cdot\text{H}_2\text{O}$ (with a small amount of anhydrous LiI phase) and 0.46 LiI –0.54 mesoporous- Al_2O_3 (pore size = 9.6 nm) composite. The crystal structure of the $\text{LiI}\cdot\text{H}_2\text{O}$ perovskite phase is shown above.

the composite prepared for 2.4 nm pore size suggest the limited stability of the new phase. In the previous investigation of LiI –mesoporous alumina composites, a pore size dependence on the lithium ionic conductivity was also observed. The conductivity increased with decreasing pore size up to 4.2 nm, and a similar decrease of the lithium ionic conductivity was observed for the smallest pore size (2.8 nm). Although systematic pore size dependence of the ^6Li NMR measurements was not performed for LiI –mesoporous alumina composite, a similarity of the peak position, -2.8 ppm, on both the LiI and Li_2ZnI_4 based composite suggests a structural similarity for the interfacial new phase. The small pore size, 2.8 nm for LiI based composite and 2.4 nm for Li_2ZnI_4 based composite, may give a restriction to the stability for the -2.8 ppm interfacial phase.

In the hydrous phase of LiI , several ionic conductor phases have been reported [16]. Among the $\text{LiI}\cdot\text{H}_2\text{O}$ system, $\text{LiI}\cdot\text{H}_2\text{O}$ with a perovskite crystal structure was shown to have highest conductivity ($7 \times 10^{-6} \text{ Scm}^{-1}$ at 298 K) [16,17]. As shown in Fig. 7, the perovskite phase, $\text{LiI}\cdot\text{H}_2\text{O}$, showed a chemical shift at -2.1 ppm. The -2.8 ppm peak apparently differs from $\text{LiI}\cdot\text{H}_2\text{O}$ phase, but the similarity of the chemical shift might suggest a structural similarity between the six-coordinated lithium environment in $\text{LiI}\cdot\text{H}_2\text{O}$ phase and the new interfacial

phase. This phase may accept higher concentration of defect in its structure, and this might be a reason of a high conductivity of this phase.

4. Conclusions

Enhancement of the electrical conductivity for Li_2ZnI_4 –mesoporous Al_2O_3 was observed. More than fifty times enhancement of the electrical conductivity was observed for the composite with mesopore size greater than 5.0 nm. ^6Li MAS-NMR measurement showed two additional lithium ion environments, at -0.8 ppm and -2.8 ppm, respectively, for the composite in both LiI and Li_2ZnI_4 . The origin of -0.8 ppm peak was suggested to be a surface lithium ion adjacent to OH^- groups. The high conductivity of the composite was suggested to originate from new interfacial component that have ^6Li NMR peak position at -2.8 ppm. Structural similarity between $\text{LiI}\cdot\text{H}_2\text{O}$ perovskite phase and the new phase was suggested.

Acknowledgment

The present work was supported in part by the Grant-in Aid for Scientific Research on Priority Area Nanoionics (439) by the Ministry of Education, Culture, Sports and Technology.

References

- [1] J. Maier, Prog. Solid State Chem. 23 (1995) 171.
- [2] H.L. Tuller, Solid State Ionics 131 (2000) 143.
- [3] C.C. Liang, J. Electrochem. Soc. 120 (1973) 1289.
- [4] F.W. Poulsen, N.H. Andersen, B. Kindl, J. Schoonman, Solid State Ionics 9/10 (1983) 119.
- [5] S. Pack, et al., J. Electrochem. Soc. 127 (1980) 2177.
- [6] A. Peters, C. Korte, D. Hesse, N. Zakharov, J. Janek, Solid State Ionics 178 (2007) 67.
- [7] D. Lubben, F.A. Modine, J. Appl. Phys. 80 (1996) 5150.
- [8] E. Schreck, K. Lauger, K. Dransfeld, Z. Phys. B. 62 (1986) 331.
- [9] H. Maekawa, R. Tanaka, T. Yamamura, MRS Symp. Proc. 756 (2003) EE6.18.1.
- [10] H. Maekawa, R. Tanaka, T. Sato, Y. Fujimaki, T. Yamamura, Solid State Ionics, 175 (2004) 281.
- [11] H. Maekawa, H.-Y. Shen, Y. Fujimaki, K. Kawata, K. Shibata, M. Kawai, T. Yamamura, MRS Symp. Proc. 835 (2005) K3.8.1.
- [12] H. Maekawa, Y. Fujimaki, H.Y. Shen, J. Kawamura, T. Yamamura, Solid State Ionics, 177 (2006) 2711.
- [13] A. Peitzner, H.D. Lutz, J.K. Cockcroft, J. Solid State Chem. 87 (1990) 463.
- [14] D. Massiot, F. Fayon, M. Capron, I. King, S. Le Calve, B. Alonso, J.-O. Durand, B. Bujoli, Z. Gan, G. Hoatson, Magn. Reson. Chem. 40 (2002) 70.
- [15] G. Ardel, D. Golodnitsky, E. Peled, Y. Wang, G. Wang, S. Bajue, S. Greenbaum, Solid State Ionics 113–115 (1998) 477.
- [16] F.W. Poulsen, Solid State Ionics, 2 (1981) 53.
- [17] F.W. Poulsen, J. Raman Spectrosc. 17 (1986) 189.